

Xuzhong Wang

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Education

Columbia University, Fu Foundation School of Engineering B.S. in Computer Science	<i>May 2028 (expected)</i>
College of William & Mary B.S. in Mathematics and Computer Science, Honors in Mathematics	<i>May 2026</i> <i>GPA: 3.99/4.0</i>

Publications & Preprints

[†] *Authors listed alphabetically (mathematics convention).* * *Equal contribution.*

1. **X. Wang**, M. Jiang, T. Nair, G. Bhusal, Y. Zhang, H. Chen. “Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning.” *Findings of EMNLP 2026*; [arXiv:2608.25100](#).
2. S. Al Jannat, Y. Li, M. He, **X. Wang**, H. Zhao, J. Sun, D. Xu, E. Xu, Y. Sun. “Akita: A High Usability Simulation Framework for Computer Architecture.” *Under submission to HPCA 2027*; [arXiv:2604.28073](#).

Research Experience

Researcher, D3i Lab, College of William & Mary <i>Advisor: Dr. Haipeng Chen, Dr. Yanfu Zhang</i>	May 2025 – Oct 2026
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- Designed and implemented a **multi-objective RL framework** to train an auxiliary retriever policy for **in-context knowledge editing**, formulating demonstration selection as a constrained MDP solved with multi-objective GRPO
- Mitigated reward hacking observed in prior work via constrained reward formulation, improving editing specificity by **10%** over existing baselines.
- Leads to paper accepted at **EMNLP Findings 2026**.

Honors Thesis, Graph Theory <i>Advisors: Dr. Daniel W. Cranston, Dr. Gexin Yu</i>	March 2025 – May 2026
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- Introduced the **Generalized Extension Lemma**, strengthening the Extension Lemma for bounding recoloring sequences from a single vertex to arbitrary subgraphs.
- Developed a **global discharging argument** that exploits global graph structure, extending beyond classical local discharging to obtain sharper density bounds.
- **Proved** that 5-list recoloring has linear diameter for $\text{mad}(G) < 3 - \varepsilon$, sharp relative to Cereceda’s conjecture; **improved** the linear bound for 6-list recoloring from 3.4 to $3.75 - \varepsilon$.

Research Assistant, Scalable Architecture Lab <i>Advisor: Yifan Sun</i>	May 2024 – November 2025
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- Implemented a **data recording tool** in Go and SQLite for simulation data tracking, enabling efficient read/write support for experiments.
- Maintained **Akita**, an **open-source** computer architecture simulator engine (**100+ GitHub stars**), improving developer experience through **event-driven design** and **smart ticking** for simulation.

Professional Experience

Machine Learning Engineer Intern, AnytimeAI

June 2026 – Aug. 2026

White Plains, NY

- Built the team’s eval harness for long-document LLM agents (deterministic citation audit, token-cost accounting, 3-pass LLM judge with Python-side aggregation); it now gates every model swap. Replacing a rubber-stamping judge moved identical outputs **11 points**.
- Benchmarked **7** agent architectures over **15** legal-corpus cases (to **16,559 pages**), catching confounds that had inflated a frontier model by **19 points**; shifted model selection to a cheap-executor config beating it by **+16.8 points at ~1/10 cost**, and cut paid vision calls **82%** via a free text-extraction tier.
- Shipped a deterministic record census after prompt fixes failed on multi-record coverage collapse, lifting the hardest case **15** → **42** in accuracy.

Machine Learning Engineer Intern, Bestlink Technology

May 2024 – July 2024

Nanjing, Jiangsu

- Designed a **data retrieval pipeline** with OpenCV to extract QA pairs from images.
- Partnered with senior engineers to architect a **LangChain-based LLM agent workflow** powering a question-answering tool for consulting clients.

Talks & Presentations

- Poster: “List-Recoloring of Graphs with Small Maximum Average Degree.” Joint Mathematics Meetings (JMM), Washington D.C., January 2026.

Grants & Awards

- Charles Center Summer Research Grant, College of William & Mary, 2025, 2026.
- Semester Research Grant, College of William & Mary, 2025.
- Cissy Patterson Grant for Research, College of William & Mary, 2025.
- JMM Travel Grant, Department of Mathematics, College of William & Mary, 2026.